Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: "IGN Video Game Reviews Dataset" (`ign.csv`)

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

Description: In this scenario, the dataset is loaded using Pandas and basic exploration is performed by viewing the first and last rows along with the dataset shape. Unnecessary columns such as `Unnamed: 0` are removed. Missing values in important columns like `score` and `genre` are handled using mean and mode respectively. Data types are also corrected to ensure proper analysis. This step ensures that the dataset is clean and ready for further processing.

Task	Description
1.	Load the `ign.csv` dataset using Pandas.
2.	Display the first 5 rows (`head()`), last 5 rows (`tail()`), and the shape of the dataset.
3.	Remove the unnecessary column: `"Unnamed: 0"`.
4.	Check for missing values in `score`, `genre`, and `platform`.
5.	Handle missing values: Fill the numeric column `score` with the mean, and the categorical column `genre` with the mode.
6.	Ensure correct data types: Convert `score` → float and `release_year`, `release_month`, `release_day` → integer.

```
First 5 rows:
    Unnamed: 0 score_phrase                                        title  ... release_year release_month  release_day
0           0      Amazing                            LittleBigPlanet PS Vita  ...         2012             9           12
1           1      Amazing  LittleBigPlanet PS Vita -- Marvel Super Hero E...  ...         2012             9           12
2           2        Great                               Splice: Tree of Life  ...         2012             9           12
3           3        Great                                            NHL 13  ...         2012             9           11
4           4        Great                                            NHL 13  ...         2012             9           11

[5 rows x 11 columns]

Last 5 rows:
        Unnamed: 0 score_phrase                                    title  ... release_year release_month  release_day
18620       18620         Good                 Tokyo Mirage Sessions #FE  ...         2016             6           29
18621       18621      Amazing          LEGO Star Wars: The Force Awakens  ...         2016             6           29
18622       18622     Mediocre  Star Ocean: Integrity and Faithlessness  ...         2016             6           28
18623       18623  Masterpiece                                   Inside  ...         2016             6           28
18624       18624  Masterpiece                                   Inside  ...         2016             6           28

[5 rows x 11 columns]
```

```python
# Display dataset info
print("First 5 rows:\n", df.head())
print("\nLast 5 rows:\n", df.tail())
print("\nShape:", df.shape)
```

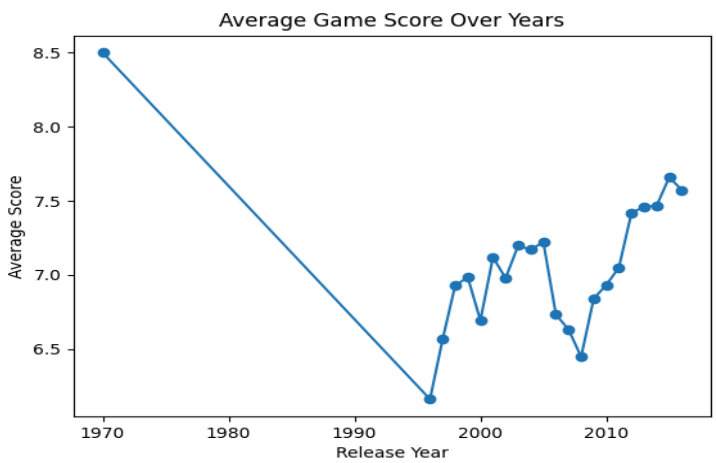
Conclusion: Data preprocessing improved the overall quality and reliability of the dataset. Missing values were handled effectively using mean and mode techniques.Unnecessary columns were removed to simplify the dataset.
Correct data types ensured accurate calculations and analysis.
This step created a strong foundation for further analysis and visualization.

Scenario 2: Line Graph (Score Trend) + Save (🟢 Easy)

Description: The line graph effectively shows how average game scores change over time. It highlights fluctuations in early years and gradual improvement in later years. The trend suggests that game quality has improved with time. Grouping data by year simplifies analysis and reveals clear patterns. Overall, this scenario helps in understanding long-term trends in the gaming industry.

Task	Description
1.	Group data by `release_year`.
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with `release_year` on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: `plt.savefig("avg_score_trend.png")`.

Graph:



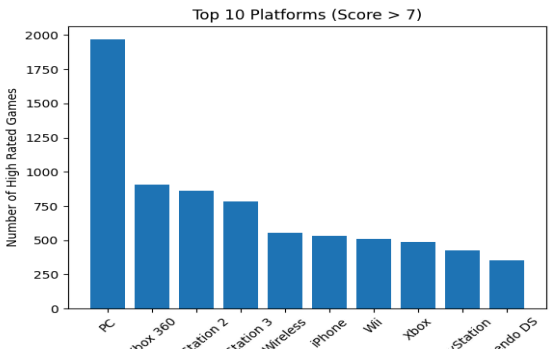
Conclusion: The line graph revealed trends in game scores over the years, helping to understand whether game quality is improving or fluctuating.

Scenario 3: Filtering + Bar Chart + Save (🟡 Medium)

Description: In this scenario, high-rated games (score greater than 7) are filtered from the dataset. The number of such games is counted for each platform. A bar chart is created to display the top 10 platforms with the highest number of highly rated games, providing insights into platform performance.

Task	Description
1.	Filter the dataset to include only games where `score > 7`.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: `plt.savefig("top_platforms_bar.png")`.

Graph:



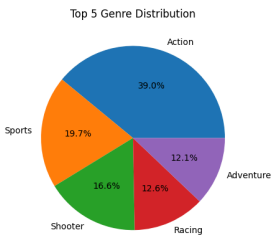
Conclusion: The bar chart clearly compared platforms based on high-rated games. It identified which platforms produce better quality games. Some platforms showed significantly higher performance than others. This helps in understanding user preference and platform popularity. Overall, it highlights key contributors in the gaming industry.

Scenario 4: Aggregation + Pie Chart + Save (🟡 Medium)

Description: The pie chart clearly represents the distribution of the top game genres in the dataset. It shows that the Action genre holds the largest share, followed by genres like Sports and Shooter, indicating their strong presence. This suggests that users generally prefer fast-paced and interactive gameplay experiences. The visualization makes it easy to understand the proportion of each genre and compare their popularity. Overall, this analysis helps in identifying dominant genres and understanding market demand in the gaming industry.

Task	Description
1.	Count the total number of games per `genre`.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using `genre` as labels and `count` as values.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("genre_distribution.png")`.

Graph:



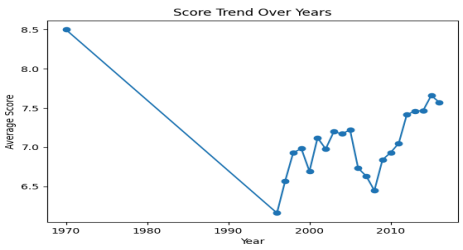
Conclusion: The pie chart demonstrated that certain genres dominate the gaming industry. The Action genre, in particular, holds the largest share, indicating strong user preference.

Scenario 5: Advanced Analysis + Multiple Graphs (🔴 Hard)

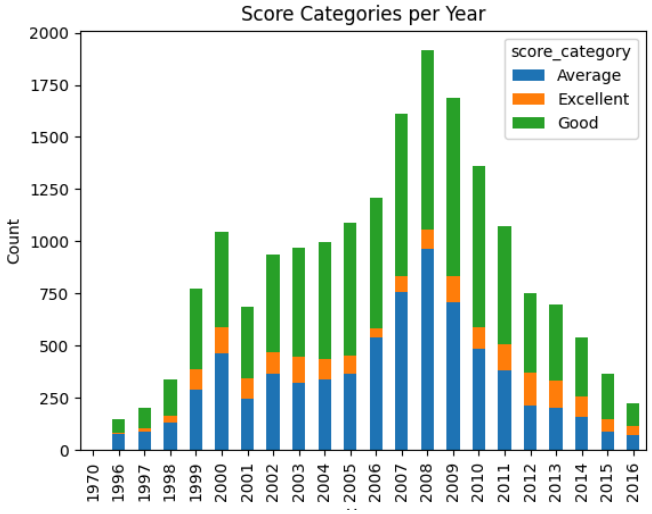
Description: This scenario focuses on advanced analysis using feature engineering and statistical methods. A new column `score_category` classifies games into Excellent, Good, and Average for better understanding. The `editors_choice` column is converted into numerical form to analyze its relationship with scores.Visualizations like line graphs, stacked bar charts, and histograms help reveal trends and distributions.Overall, this scenario provides deeper insights into score patterns and relationships in the dataset.

Task	Description
1.	Create a new column `score_category`: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2.	Convert the `editors_choice` column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3.	Use NumPy's `np.diff()` to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per `release_year`.
5.	Plot a **Stacked Bar Chart** showing the count of `score_category` per `release_year`.
6.	Plot a **Histogram** showing the distribution of `score`.
7.	Save the graphs: `plt.savefig("score_trend.png")`, `plt.savefig("score_category_stacked.png")`, `plt.savefig("score_distribution.png")`.
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if `editors_choice` correlates with high scores.

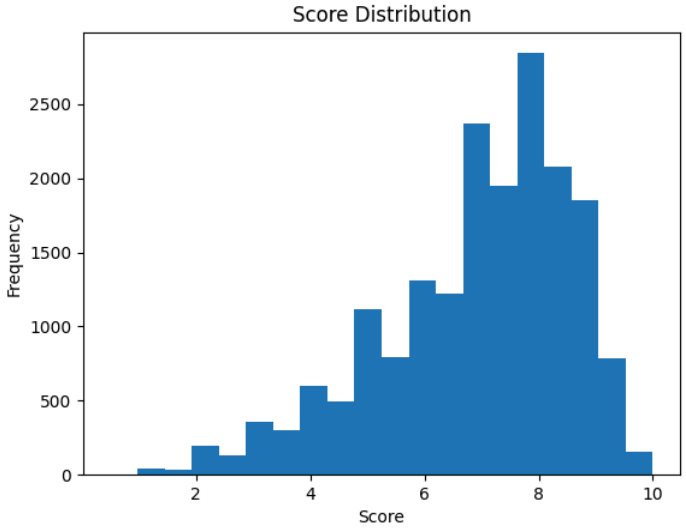
Graph 1 :- Line Graph



Graph 2:-



Graph 3:-



Conclusion: Advanced analysis provided deeper insights into score patterns, category distributions, and statistical relationships. The use of feature engineering and multiple visualizations helped in understanding the dataset more effectively.

Final Conclusion: By combining data preprocessing, visualization, and advanced analysis, this project successfully extracts meaningful insights from the dataset. Each scenario contributes to a different aspect of analysis, ultimately providing a comprehensive understanding of trends, patterns, and relationships in the gaming data.

What you have learned :- In this project, I learned how to perform complete data analysis starting from data loading and preprocessing to advanced analysis and visualization. I understood how to clean data by handling missing values, removing unnecessary columns, and correcting data types. I learned to use different types of graphs like line charts, bar charts, pie charts, and histograms to analyze trends, comparisons, and distributions. I also gained knowledge of feature engineering and converting categorical data into numerical form for deeper analysis. Overall, I learned how to extract meaningful insights and make data-driven interpretations from a real-world dataset.